Google Developer Group on Campus
CAMBRIDGE INSTITUTE OF TECHNOLOGY

Newsletter

January Edition

THE LEADS DESK

- **At GDGoC-CIT, Our Intent Has Always Been Clear:** To Build An Ecosystem Where Students Don't Just Learn Technology, But Learn How Technology Works In The Real World. We Focus On Creating Exposure To Hackathons, Industry- Aligned Technical Events, And Project-Based Learning That Mirrors The Expectations Of Today's Tech Landscape. Every Initiative We Take Is Designed To Bridge The Gap Between Classroom Knowledge And Industry Application.

- One Strong Learning From This Month Has Been Seeing How Quickly Students Adapt When Given The Right Environment. Through Our Recent Engagements, It Became Evident That When Students Are Exposed To Structured Problem Statements, Real Constraints, And Collaborative Workflows, Their Approach To Technology Transforms. They Stop Thinking In Terms Of Marks And Start Thinking In Terms Of Impact, Scalability, And Execution.

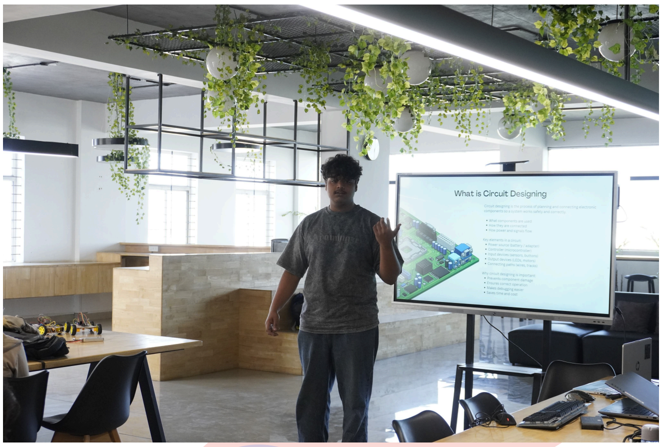
- Looking Ahead, **GDGoC-CIT Will Continue To Push Boundaries, Bringing In More Industry, Driven Sessions, Hands-On Challenges, And Platforms Where Ideas Are Tested, Refined, And Shipped.** Our Focus Will Remain On Nurturing Projects That Meet Real-World Criteria And Preparing Students To Confidently Step Into Professional Tech Spaces.

- To The GDGoC-CIT Community: Stay Curious, Stay Bold, And Keep Building. This Is Your Space To Experiment, Fail Fast, And Grow Stronger—Together.

THIS MONTH AT GDG

- **The GDG X Ad Astra Collaborative Workshop On Embedded Systems** Was An Immersive And Inspiring Experience That Provided Valuable Insights Into The Fundamentals Of Embedded Systems And Their Applications In Solving Real-Life Problems. The Session Effectively Bridged Theory And Practice, Helping Participants Understand How Embedded Technology Plays A Crucial Role In Everyday Innovations. What Made The Workshop Especially Engaging Was Its Interactive Nature, Led By **The Club Leads, Shreya Upadhyay (GDG) And S Nishaath (Ad Astra), Along With The**

Embedded Systems Leads, Sankarshan V Sastry (GDG) And Yashas Mahesh (Ad Astra), Who Encouraged Open Discussion And Active Participation Throughout The Session. The Curiosity, Enthusiasm, And Energy In The Room Were Truly Remarkable, With Everyone Driven By A Shared Passion To Learn, Explore, And Create Meaningful Solutions. A Major Highlight Of The Workshop Was The Hands-On Session, Where Participants Were Introduced To Simulation Tools Used For Designing Embedded Systems, Making The Learning Process Both Practical And Exciting. Overall, The Workshop Was A Wellstructured And Impactful Learning Experience That Not Only Strengthened Technical Understanding But Also Inspired Participants To Think Innovatively And Work Towards Real-World Breakthroughs.



← *Explanation of components of embedded systems*

Introduction to Embedded Systems →

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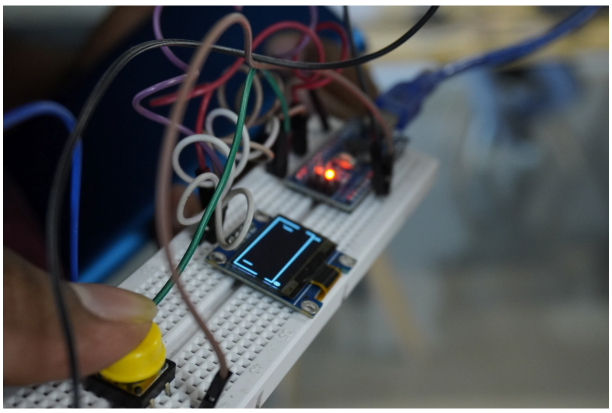
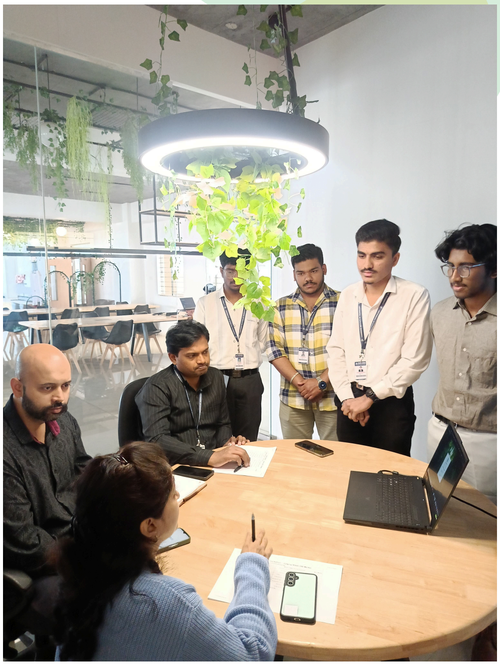


- **PulseX, An Open-Innovation Techfest Hosted By GDGoC CIT** Was Designed To Encourage Students To Build Impactful Solutions For Real-World And Social Challenges Using Google Technologies. It Provided A Platform For Participants To Ideate, Experiment, And Transform Ideas Into Practical, Working Products While Showcasing Their Technical And Creative Skills. PulseX Followed A Clear And Welldefined Three-Phase Structure That Guided Teams From Idea Conception To Complete Execution. **Phase 1, The Ideathon,** Focused Solely On Idea Pitching, Allowing Teams To Concentrate On Identifying Real-World Problems, Proposing Relevant Use Cases, And Presenting Innovative Yet Feasible Solutions. Teams Were Shortlisted Based On The Clarity, Impact, And Practicality Of Their Ideas. **Phase 2, The Partial Prototype Phase,** Required Teams To Present A Basic Or Partially Working Prototype, Where Evaluation Was According To The Technical Approach, Solution Architecture, Feasibility, And Clear Justification Of Google Technology Usage Rather Than Full Functionality. **Phase 3, The Final Round,** Showcased Fully Working End-To-End Prototypes And Used A Detailed 100-Mark Evaluation Framework Assessing Core Functionality, Live Demo Quality, UI/UX, Effective Integration Of Google Technologies, Technical Depth, Scalability, Presentation Clarity, And Q&A Performance. Overall, The Event Followed A Progressive Flow From Ideation To Execution, Prioritizing Real-World Impact, Usability, And Functional Solutions Over Complexity And Became A Grand Success.



LEARNING CORNER: VERTEX AI STUDIO

- Ever Wondered How People Experiment With AI Models Without Writing Pages Of Complex Code? Vertex AI Studio Is Google's Quiet Powerhouse That Makes This Possible. It's A Beginner-Friendly Tool Inside Google Cloud That Lets You Explore, Test, And Build AI Models Through A Clean, Interactive Interface.

- Vertex AI Studio Allows You To Prompt, Test, And Fine-Tune Generative AI Models Using Text, Images, Or Your Own Data. You Can Compare Responses, Refine Prompts, And Move From Idea To Prototype Quickly.

Why Is It Useful?

It Removes The Heavy Lifting Of Infrastructure, Scaling, And Model Management. Built-In Monitoring, Safety Controls, And Version Tracking Make Experimentation Safe And Organized, Even For Beginners. You Learn Faster Because The Platform Handles The Technical Complexity.

How Can You Access It?

Simply Open The Google Cloud Console And Navigate To Vertex AI - Studio.

Who Should Explore It?

Students, Beginners, Developers, Startups, And Teams Curious About Applying AI To Real Problems.

What's It An Alternative To?

Tools Like OpenAI Playground. Vertex AI Studio Stands Out With Seamless Google Cloud Integration, Better Data Control, And Enterprise-Grade Security.

Example: A Student Creating An AI Study Helper Can Test Prompts, Improve Answers, And Scale It Later As An App, All In One Place.

WHAT'S NEW IN TECH

1). India Goes Big On AI, Government Of India Has Launched A Public AI Infrastructure Strategy, Deploying Large-Scale Compute Including NVIDIA H100 GPUs, Low-Cost Access Via The India AI Portal, Shared Datasets Through AI Kosha, And Edge AI Servers Across Railways And Hospitals, Offering Compute At Significantly Lower Costs Than Global Providers Like AWS And Google Cloud, With Free Or Subsidized Access For Students And Startups, Thereby Lowering Barriers To AI Research And Positioning India As A Serious Global AI Player.

2). Google Expands Personalized AI With Gemini, Integrating Gmail, Photos, And Search To Deliver Context-Aware Responses Based On User Data, Highlighting A Shift From Generic AI Answers To Highly Personalized Intelligence.

3). Google And Apple Sign A Reported $21B AI Partnership, Enabling Google's Models To Power Siri Across Apple Devices, Potentially Extending Google AI Reach To ~4.6 Billion Devices And Raising Concerns Around Concentration Of AI Control.

4). Anthropic Introduces Claude Co-Work, Allowing Claude To Directly Act On Folders And Files To Complete Tasks Such As Document Organization And Meeting Preparation, Marking Progress Toward AI Agents That Execute Work Rather Than Only Respond.

5). Pixverse R1 Introduces Real-Time AI Video Generation, Enabling Instant Video Creation From Text Prompts Without Traditional Rendering Delays, Signaling Faster Iteration In Generative Video Tools.

WHAT'S COMING UP
GOOGLE SUMMER OF CODE (GSOC) 2026

Google Summer Of Code Is A Global, Online Program Focused On Bringing New Contributors Into Open Source Software Development. GSoC Contributors Work With An Open Source Organization On A 12+ Week Programming Project Under The Guidance.

Contributors Join Mentoring Organizations, Agree On A Project Proposal, And Spend The Summer Coding Under Mentor Guidance. All The Work Is Done Online And Culminates In Code That Often Gets Merged Into Widely-Used Open-Source Projects.

Why Participate In Google Summer Of Code 2026?

Real-World Experience Through Contributions To Production-Level Open-Source Projects
Mentorship From Experienced Open-Source Maintainers
Strong Resume Value With Global Recognition
Exposure To A Global Developer Community
Development Of Industry-Ready Skills

Timeline

Feb 19 – Mar 15: Community Discussions With Mentoring Organizations
Mar 16 – Mar 31: Contributor Application Period
Apr 30: Accepted Contributor Projects Announced
May 1 – May 24: Community Bonding Period
May 25: Coding Officially Begins
July 6 – July 10: Midterm Evaluations
Aug 17 – Aug 24: Final Week And Contributor Evaluations
Aug 24 – Aug 31: Mentor Evaluations
Nov 2: Extended Timeline Coding Deadline (If Applicable)
Nov 9: Final Results Announced (Extended Projects)

Common FAQ

Who Can Apply?

Anyone Aged 18 Or Older Can Apply — Students And Non-Students Alike. You Don't Need To Be In A Specific College Year Or Degree Programme.

Do I Need Open-Source Experience?

No Prior Open-Source Experience Is Required, But Having Some GitHub Contributions Or Engaging With A Project Community Before Applying Improves Your Chances Significantly.

What Do I Earn?

Selected Contributors Receive A Stipend From Google For Successfully Completing Their Project. The Exact Amount Is Announced Yearly And May Vary By Location.

How Long Is The Program?

The Coding Project Period Lasts 12+ Weeks, And In Some Cases Extended By The Mentoring Organization.

What Skills Do I Need?

The Exact Requirements Depend On The Project You Choose. Common Skills Include Programming Languages, Git/GitHub Usage, Communication, And Eagerness To Learn.

USEFUL REFERENCE LINKS

Official Site: https://summerofcode.withgoogle.com/t
Timeline: your paragraph texthttps://developers.google.com/open-source/gsoc/timeline how it Works: your paragraph texthttps://summerofcode.withgoogle.com/how-it-works

STAY CONNECTED.....

Be Part Of Our Growing Community And Stay In The Loop With Updates On Events, Opportunities, Resources, And Announcements. Explore All Our Community Platforms Here: texthttps://linktr.ee/gdg_citech